

# Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

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### 3. Supplemental eMethods

#### A. Inclusion Criteria

For inclusion in the study, participants fulfilled the following criteria:

- Severe aortic stenosis (aortic valve peak velocity  $\geq 4.0$  m/s, or aortic valve area indexed to body surface area  $< 0.6 \text{ cm}^2/\text{m}^2$  with aortic valve peak velocity  $\geq 3.5$  m/s)
- Age over 18 years
- No symptoms attributable to aortic stenosis that require aortic valve replacement

#### B. Exclusion Criteria

Participants were not enrolled into the study if any of the following criteria were fulfilled:

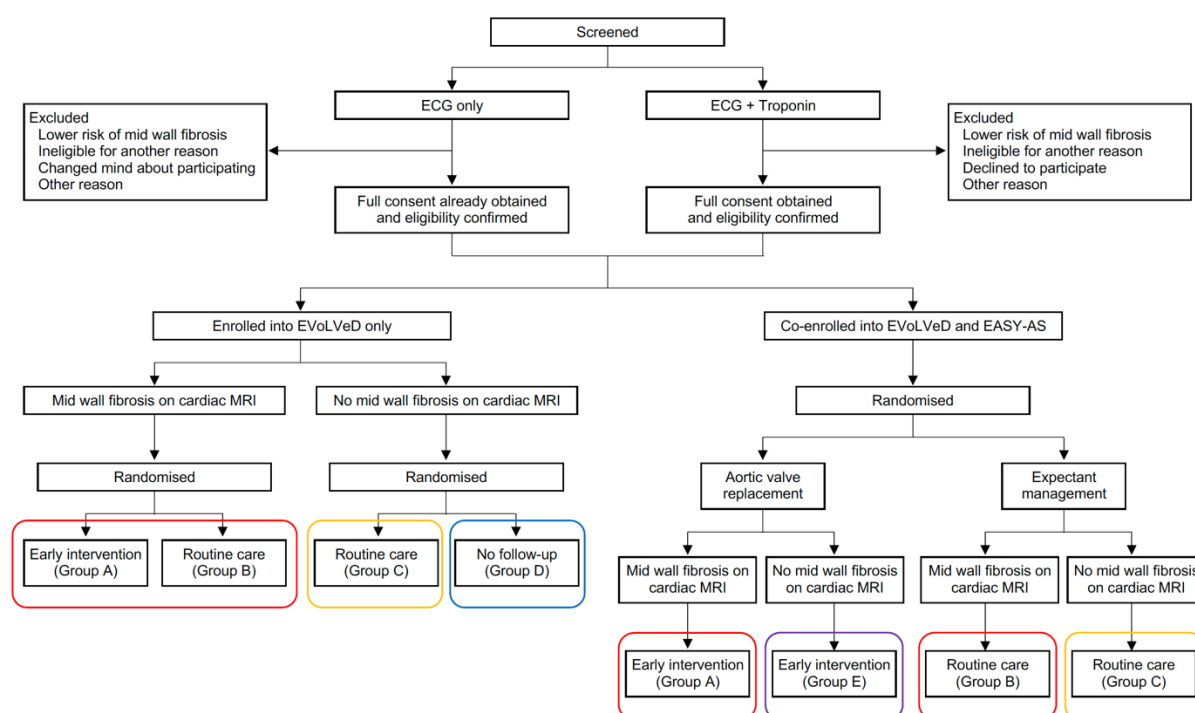
- Deemed lower risk for midwall fibrosis on screening
- Planned cardiac surgery
- Previous valve replacement
- Severe hypertension (systolic  $> 180$  or diastolic  $> 110$  mmHg)
- Acute pulmonary oedema or cardiogenic shock
- Left ventricular ejection fraction  $< 50\%$  on cardiac MRI
- Significant abnormalities on cardiac MRI that would prevent enrolment
- Coexistent severe aortic regurgitation or mitral regurgitation
- Coexistent mitral stenosis greater than mild in severity
- Coexistent hypertrophic cardiomyopathy or cardiac amyloidosis
- Any contraindication to MRI scanning (such as permanent pacemaker)
- Advanced renal impairment (glomerular filtration rate  $< 30 \text{ mL/min/1.73 m}^2$ )
- Pregnancy or breast feeding
- Patient judged to be unfit to be considered for aortic valve replacement or transcatheter aortic valve implantation
- Patient declines to consider undergoing valve replacement surgery or transcatheter aortic valve implantation
- Inability to give informed consent
- Previous randomization into this study

#### C. Cardiac Magnetic Resonance Analysis

The presence of midwall late gadolinium enhancement as a marker of myocardial fibrosis was independently assessed by the core laboratory blinded to clinical details. First, the core laboratory assessed each scan for acceptable image quality according to standardized criteria. The presence or absence of midwall late gadolinium enhancement was assessed qualitatively by two experienced readers (MD and RE, RB, KL or NC), and differentiated from the pattern of scarring due to prior myocardial infarction, based upon short axis slices in two-phase encoding directions and long-axis images (Supplement eTable 2). The opinion of a third independent experienced reader was sought for any discordant classifications, and site investigators and attending physicians were blinded to the cardiac magnetic resonance findings other than any unexpected clinically urgent findings.

#### D. Details of co-enrolment with the EASY-AS Trial

During the recruitment phase of the trial, the EASY-AS trial opened to recruitment, randomizing patients with asymptomatic severe aortic stenosis but not necessarily with myocardial fibrosis, to early aortic valve intervention or guideline-directed conservative management (NCT04204915). The EVOLVED trial permitted co-enrolment with the EASY-AS trial. Patients from EASY-AS who were eligible for EVOLVED took the trial arm allocation from the EASY-AS randomization. The flow of patients through EVOLVED and EASY-AS is shown on the following figure. In total, 47 patients were co-enrolled in the EVOLVED and EASY-AS trials.



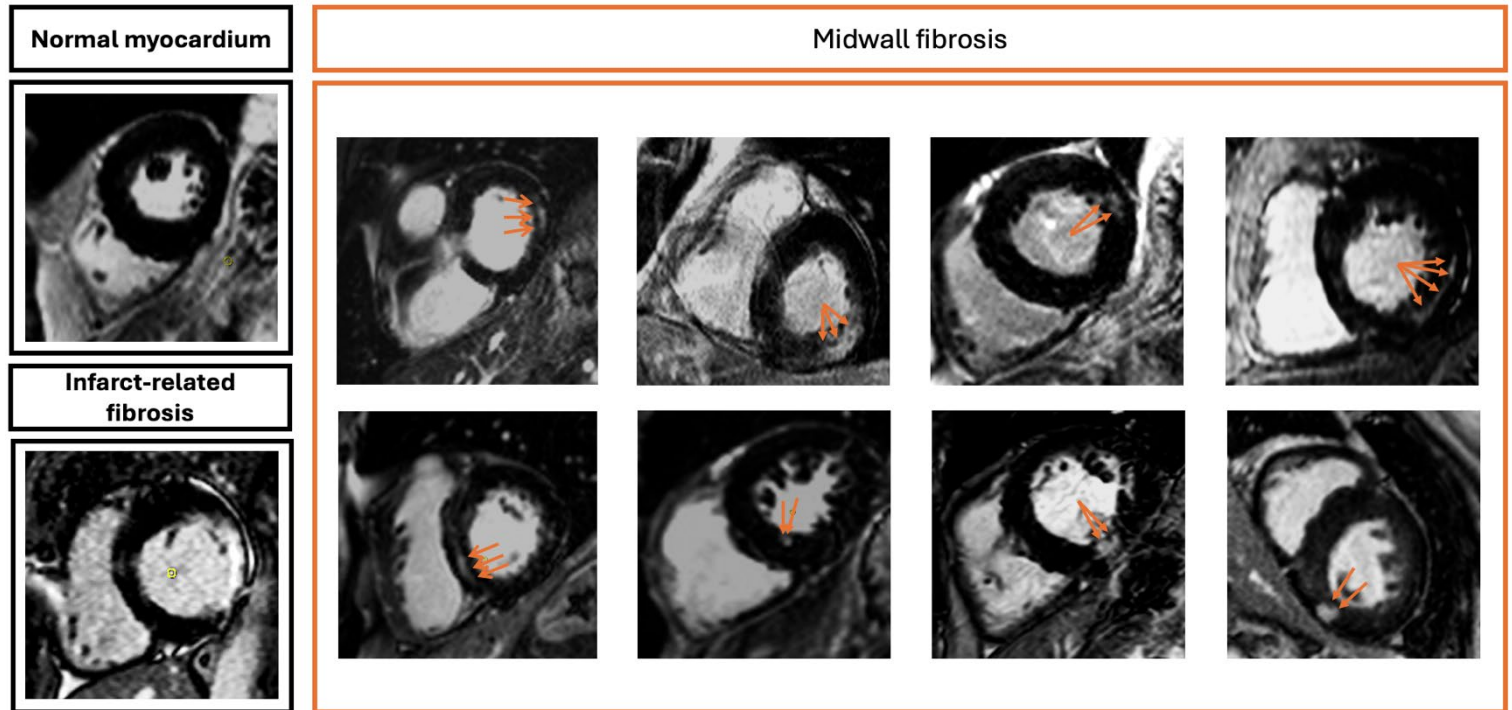
## E. Details of patients with midwall fibrosis withdrawn or excluded from analysis

One participant who was randomized to early aortic valve intervention was excluded from the modified intention-to-treat analysis. This patient was randomized but subsequently identified to be ineligible because of a referral for aortic valve surgery prior to being approached to participate in the trial.

A second participant randomized to guideline-directed conservative management was initially randomized in error on the test site rather than the live randomization site. Before they were randomized on the live site, their eligibility was not re-confirmed, but they had developed symptoms and had been referred by another cardiologist for aortic valve intervention. When this was subsequently identified, the patient was withdrawn by the Principal Investigator, without any follow-up being completed. As a result, the patient was also excluded from the modified intention-to-treat analysis.

#### 4. Supplemental Figures

##### A. eFigure 1: Evidence of Myocardial Fibrosis on Magnetic Resonance

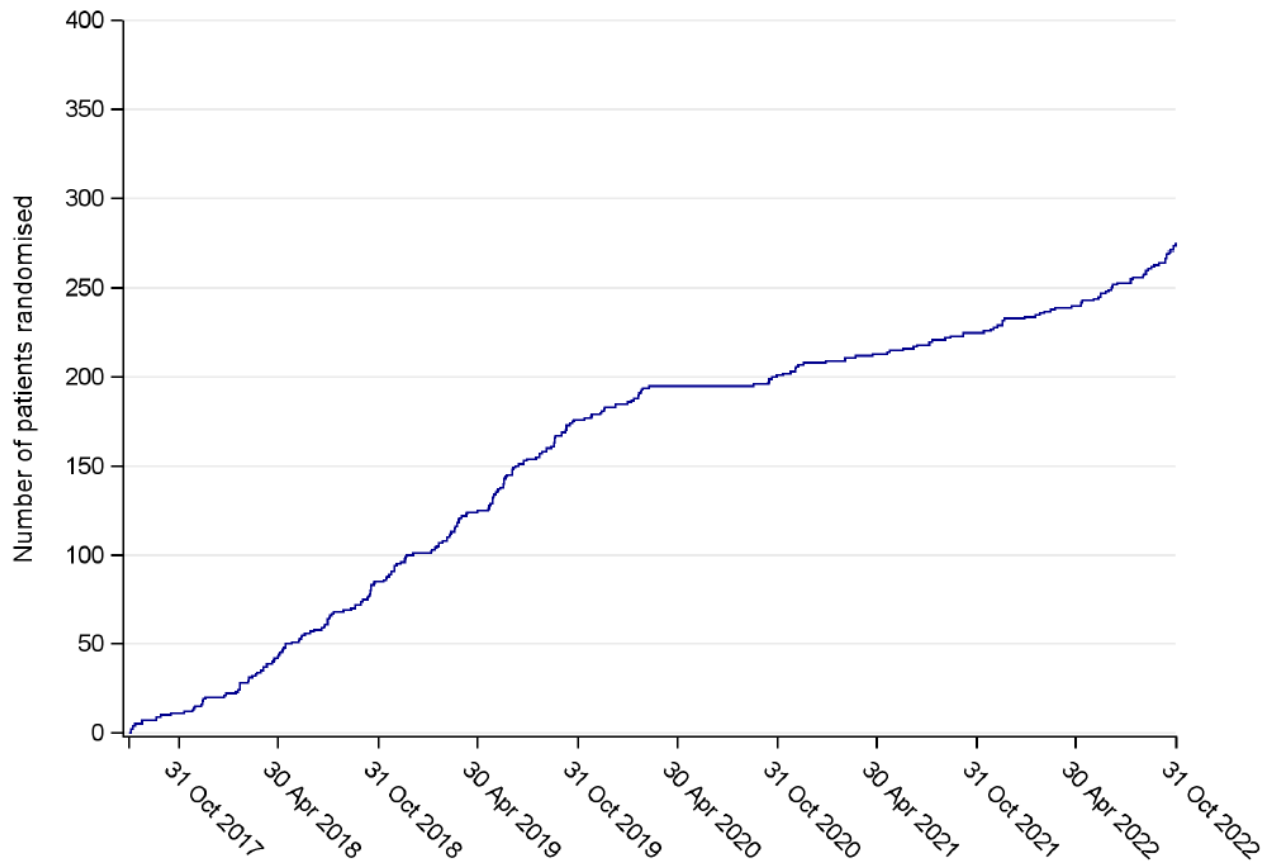


Examples of short axis slices of left ventricular myocardium with and without evidence of myocardial fibrosis on magnetic resonance, taken from participants who were screened for this study.

**Left Panel:** Upper; There is no late gadolinium enhancement reflecting an absence of myocardial fibrosis. Lower; there is a myocardial infarction pattern of late gadolinium enhancement. Neither of these patients met the criteria for randomization to early intervention or guideline-directed conservative management and were instead entered into an observational cohort to maintain blinding of the cardiac findings to the treating physician.

**Right Panel:** Myocardium with midwall late gadolinium enhancement depicted by the orange arrows. These patients met inclusion criteria for the trial and were randomized.

## B. eFigure 2: Trial Recruitment



*First patient was randomised on 04-AUG-2017 and last patient was randomised on 31-OCT-2022.*

Recruitment to the trial was halted on 17 March 2020 due to the COVID-19 pandemic in line with a United Kingdom Government directive. Recruitment to the trial did not fully recover, and the results of the AVATAR and RECOVERY trials suggested a larger treatment effect of early intervention than had been assumed in the original power calculation. A decision to halt recruitment into the trial on the pre-specified date of 31 October 2022 was therefore taken by the Trial Steering Committee.

## **B     eFigure 3: New York Heart Association Symptom Classification**

At one-year follow up, patients allocated to early intervention had a lower burden of symptoms as assessed by the New York Heart Association classification compared to those allocated to guideline-directed conservative management: adjusted odds ratio 0.37 [95% confidence interval 0.20 to 0.70].

## 5. Supplemental Tables

**A. eTable 1: Cardiac Magnetic Resonance Imaging Protocol**

|                                                                                                     | Number of acquisitions | Estimated time (mins) | Time post-gadolinium |
|-----------------------------------------------------------------------------------------------------|------------------------|-----------------------|----------------------|
| Subject positioned within scanner, ensuring high quality ECG signal and intravenous cannula patency |                        | 5                     |                      |
| <b>Sequence</b>                                                                                     |                        |                       |                      |
| Localiser                                                                                           | 1                      |                       |                      |
| Site specific method of localisers to derive appropriate cardiac planes                             | 1-5                    | 1-3                   |                      |
| 4 chamber (horizontal long axis) cine                                                               | 1                      | 2                     |                      |
| 2 chamber (vertical long axis) cine                                                                 | 1                      | 2                     |                      |
| 3 chamber cine                                                                                      | 1                      | 2                     |                      |
| LV short axis stack cine- 8mm slice thickness, no gap                                               | 14-15                  | 6                     |                      |
| MoLLI (basal and mid ventricle slices)-<br><i>Optional</i>                                          | 1                      | 2                     |                      |
| <b>Gadolinium injection (0.15 mmol/kg)</b>                                                          |                        |                       |                      |
| Single scout- any image, localiser                                                                  | 1                      | 1                     | 0-1                  |
| LVOT cine (orthogonal to 3 chamber)                                                                 | 1                      | 2                     | 1-3                  |
| Aortic valve FLASH                                                                                  | 1                      | 2                     | 3-5                  |
| Break- 7 minutes post gadolinium injection                                                          |                        | 0-2                   | 5-7                  |
| T1 scout- To determine optimal T1                                                                   | 1                      | 1                     | 7-8                  |
| GRE-T1 LGE LV short axis stack- 8mm slice thickness, no gap                                         | 14-15                  | 7                     | 15-22                |
| Phase swap GRE-T1 LGE LV short axis stack                                                           | 14-15                  | 7                     | 15-22                |
| Long axis GRE-T1 LGE images                                                                         | 3                      | 3                     | 24-27                |
| MOLLI (basal and mid ventricle slices)-<br><i>Optional</i>                                          | 1                      | 2                     | 22-24                |

ECG: Electrocardiogram; GRE-T1: Gradient echo sequence- T1; LGE: Late gadolinium enhancement; LV: Left ventricle; LVOT: Left ventricular outflow tract; MOLLI: Modified Look-Locker Inversion recovery;

**B. eTable 2: Cardiac Magnetic Resonance Reporting Protocol**

|                                                                                                                                                                                                                                  |                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Image quality</b>                                                                                                                                                                                                             |                         |
| Adequate                                                                                                                                                                                                                         | Y/N                     |
| <b>Aortic valve anatomy</b>                                                                                                                                                                                                      |                         |
| Bicuspid valve present                                                                                                                                                                                                           | Y/N                     |
| <b>Left ventricular volumetric analysis</b>                                                                                                                                                                                      |                         |
| Indexed left ventricular end diastolic volume (ml/m <sup>2</sup> )                                                                                                                                                               |                         |
| Indexed left ventricular end systolic volume (ml/m <sup>2</sup> )                                                                                                                                                                |                         |
| Indexed LV stroke volume (ml/m <sup>2</sup> )                                                                                                                                                                                    |                         |
| Left ventricular ejection fraction (%)                                                                                                                                                                                           |                         |
| Indexed left ventricular mass (g/m <sup>2</sup> )                                                                                                                                                                                |                         |
| <b>Atrial analysis</b>                                                                                                                                                                                                           |                         |
| Indexed biplanar left atrial volume (ml/m <sup>2</sup> )                                                                                                                                                                         |                         |
| <b>Qualitative analysis of late gadolinium enhancement</b>                                                                                                                                                                       |                         |
| Midwall late gadolinium enhancement present                                                                                                                                                                                      | Y/N                     |
| If yes, AHA segments                                                                                                                                                                                                             | Focal/Linear/ Patchy    |
| Infarct pattern late gadolinium enhancement present                                                                                                                                                                              | Y/N                     |
| If yes, AHA segments                                                                                                                                                                                                             |                         |
| <b>Quantitative analysis of late gadolinium enhancement</b>                                                                                                                                                                      |                         |
| Midwall late gadolinium enhancement (g)                                                                                                                                                                                          |                         |
| Midwall late gadolinium enhancement (% of myocardium)                                                                                                                                                                            |                         |
| Infarct pattern late gadolinium enhancement (g)                                                                                                                                                                                  |                         |
| Infarct pattern late gadolinium enhancement (% of myocardium)                                                                                                                                                                    |                         |
| <b>Cardiac magnetic resonance eligibility check</b>                                                                                                                                                                              |                         |
| Evidence of co-existent cardiomyopathy                                                                                                                                                                                           | Y/N                     |
| Any cardiac MRI features that would prevent enrolment?                                                                                                                                                                           | Y/N                     |
| Can patient proceed to randomization?                                                                                                                                                                                            | Y/N                     |
| <b>Native T1 mapping by AHA segments (ms)</b>                                                                                                                                                                                    |                         |
| 1                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 2                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 3                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 4                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 5                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 6                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 7                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 8                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 9                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 10                                                                                                                                                                                                                               | Infarct LGE present Y/N |
| 11                                                                                                                                                                                                                               | Infarct LGE present Y/N |
| 12                                                                                                                                                                                                                               | Infarct LGE present Y/N |
| <b>Post-contrast T1 mapping by AHA segments (ms)</b>                                                                                                                                                                             |                         |
| 1                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 2                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 3                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 4                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 5                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 6                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 7                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 8                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 9                                                                                                                                                                                                                                | Infarct LGE present Y/N |
| 10                                                                                                                                                                                                                               | Infarct LGE present Y/N |
| 11                                                                                                                                                                                                                               | Infarct LGE present Y/N |
| 12                                                                                                                                                                                                                               | Infarct LGE present Y/N |
| <b>ECV (%) using the formula below:</b>                                                                                                                                                                                          |                         |
| - ECV% = partition coefficient x [1-hematocrit] x 100<br>- Partition coefficient = [ $\Delta R1_{\text{myocardium}} / \Delta R1_{\text{blood-pool}}$ ]<br>- $\Delta R1 = (1 / \text{post-contrast T1}) - (1 / \text{native T1})$ |                         |

|                                                                |  |
|----------------------------------------------------------------|--|
| <b>Indexed ECV (ml/m<sup>2</sup>) using the formula below:</b> |  |
| iECV = (ECV% x indexed LV mass) / (myocardial density x 100)   |  |
| Myocardial density = 1.05 g/mL                                 |  |

**C. eTable 3: Aortic Valve Intervention Procedures and Results**

|                                                                       | <b>Early<br/>Intervention<br/>(n=113)</b> | <b>Conservative<br/>Management<br/>(n=111)</b> |
|-----------------------------------------------------------------------|-------------------------------------------|------------------------------------------------|
| Did not receive intervention during follow-up, No. (%)                | 7 (6)                                     | 26 (23)                                        |
| Type of surgery, No. (%)                                              |                                           |                                                |
| - Elective surgery                                                    | 103 (97)                                  | 72 (85)                                        |
| - Urgent inpatient surgery                                            | 3 (3)                                     | 13 (15)                                        |
| Surgical Aortic Valve Replacement, No. (%)                            | 80 (75)                                   | 47 (55)                                        |
| - Biological prosthesis                                               | 73 (91)                                   | 44 (94)                                        |
| - Mechanical prosthesis                                               | 7 (9)                                     | 3 (6)                                          |
| Transcatheter aortic valve intervention, No. (%)                      | 26 (25)                                   | 38 (45)                                        |
| - Transfemoral                                                        | 25 (96)                                   | 37 (97)                                        |
| - Other                                                               | 1 (4)                                     | 1 (3)                                          |
| Concomitant Procedure, No. (%)                                        |                                           |                                                |
| - Coronary artery bypass grafting                                     | 20 (19)                                   | 13 (15)                                        |
| - Aortic root repair or replacement                                   | 6 (6)                                     | 4 (5)                                          |
| Primary indication for aortic valve intervention, No. (%)             |                                           |                                                |
| - Symptom development                                                 | N/A                                       | 61 (72)                                        |
| - Reduction in ejection fraction                                      | N/A                                       | 1 (1)                                          |
| - Abnormal exercise test                                              | N/A                                       | 1 (1)                                          |
| - Rapid progression of aortic stenosis                                | N/A                                       | 22 (26)                                        |
| Periprocedural complications within 30 days, No. (%)                  |                                           |                                                |
| - Death caused by immediate complication of aortic valve intervention | 1 (1)                                     | 0 (0)                                          |
| - Stroke                                                              | 1 (1)                                     | 4 (5)                                          |
| - Myocardial Infarction                                               | 0 (0)                                     | 0 (0)                                          |
| - Life-threatening or disabling bleeding                              | 3 (3)                                     | 1 (1)                                          |
| - Major vascular complication                                         | 2 (2)                                     | 2 (2)                                          |
| - Permanent pacemaker implantation                                    | 5 (5)                                     | 2 (2)                                          |
| - Endocarditis                                                        | 0 (0)                                     | 0 (0)                                          |
| - Reoperation for any reason                                          | 4 (4)                                     | 2 (2)                                          |
| - Wound dehiscence                                                    | 2 (2)                                     | 2 (2)                                          |
| - Still requiring renal replacement therapy at 30 days                | 0 (0)                                     | 1 (1)                                          |

**D. eTable 4: Causes of Death**

|                                                         | Early Intervention<br>(n=113) | Conservative<br>Management<br>(n=111) |
|---------------------------------------------------------|-------------------------------|---------------------------------------|
| <b>Total number of deaths during follow-up, No. (%)</b> | 16 (14)                       | 14 (13)                               |
| <b>Cardiovascular Death, No. (%)</b>                    | 10 (9)                        | 8 (7)                                 |
| - Aortic-stenosis related death                         | 6 (5)                         | 5 (5)                                 |
| <b>Non-cardiovascular Death, No. (%)</b>                |                               |                                       |
| - Lower respiratory tract infection                     | 3 (3)                         | 1 (1)                                 |
| - Malignancy                                            | 2 (2)                         | 2 (2)                                 |
| - Old Age                                               | 1 (1)                         | 0 (0)                                 |
| - Intracerebral haemorrhage                             | 0 (0)                         | 1 (1)                                 |
| - Intra-abdominal sepsis                                | 0 (0)                         | 1 (1)                                 |
| - Chronic lung disease                                  | 0 (0)                         | 1 (1)                                 |

**E. eTable 5: Participant Recruitment by Trial Site**

| <b>Trial Site</b>                                         | <b>No. (%)</b> |
|-----------------------------------------------------------|----------------|
| University of Edinburgh                                   | 76 (28)        |
| Northumbria Healthcare NHS Foundation Trust               | 36 (13)        |
| Guys & St. Thomas's NHS Foundation Trust                  | 26 (10)        |
| University Hospitals of Birmingham NHS Trust              | 21 (8)         |
| University of Leicester Hospitals NHS Trust               | 18 (7)         |
| Leeds Teaching Hospitals NHS Trust                        | 15 (6)         |
| Liverpool Heart and Chest Hospital NHS Foundation Trust   | 12 (4)         |
| University of Dundee and NHS Tayside                      | 7 (3)          |
| NHS Highlands                                             | 7 (3)          |
| Bristol Heart Institute                                   | 7 (3)          |
| Norfolk and Norwich University Hospitals NHS Trust        | 6 (2)          |
| University of Aberdeen and NHS Grampian                   | 5 (2)          |
| Newcastle Upon Tyne Hospitals NHS Foundation Trust        | 5 (2)          |
| West Hertfordshire Hospitals NHS Trust                    | 5 (2)          |
| Barts Health NHS Trust                                    | 5 (2)          |
| Essex Cardiothoracic Centre, Basildon University Hospital | 5 (2)          |
| Manchester University NHS Foundation Trust (Central)      | 5 (2)          |
| Royal Wolverhampton NHS Trust                             | 5 (2)          |
| Manchester University NHS Foundation Trust (Wythenshawe)  | 3 (1)          |
| Royal Papworth Hospital NHS Foundation Trust              | 2 (1)          |
| NHS Borders                                               | 1 (0)          |
| University of Glasgow and NHS Greater Glasgow and Clyde   | 1 (0)          |
| South Tees Hospitals NHS Foundation Trust                 | 1 (0)          |
| Royal Perth Hospital                                      | 1 (0)          |